Moishezon space and Moishezon morphism Summer 2025

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1 Overview

The aim of this note is to give a brief introduction to Moishezon variety and Moishezon morphism. The major references for this note are [Kol22], [Fuj83], and [Uen75].

Why study Moishezon morphisms? First, Moishezon spaces have more functorial behavior (compared with projective varieties), as we will see in Section 2. Secondly, from almost any projective variety we can construct a Moishezon space via bimeromorphic modification, making Moishezon spaces versatile in birational geometry. Thirdly, by Artin's fundamental theorem, the category of Moishezon spaces appears naturally in moduli theory. Another compelling reason to consider the Moishezon category is that it allows cut-and-paste operations similar to those we can perform in topology.

This series of talks is organized as follows:

Lec 1. Basic knowledge about Moishezon spaces and Moishezon morphisms,

Lec 2. Fiberwise bimeromorphic problems.

Lec 3. General type locus, Moishezon locus, and projective locus.

Lec 4. Projectivity criteria and behavior of projective locus.

Lec 5. Rational curves on Moishezon spaces.

Contents

2 Moishezon spaces

Definition 2.1 (Meromorphic S-map)**.** Let X, Y be reduced complex spaces. We call the S-map a *meromorphic S-map* if

$$\begin{tikzcd}
 & \Gamma \arrow[dl, "p"'] \arrow[dr, "q"] & \\
X \arrow[rr, dashed, "\alpha"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\
 & S &
\end{tikzcd}$$

the natural projection associated to the graph $p : \Gamma \to X$ is a proper bimeromorphic morphism. Moreover, if the natural projection $q : \Gamma \to Y$ is also a proper bimeromorphic morphism, then we call α a *proper bimeromorphic S-map*.

Remark 2.2 (Comparison between meromorphic map and S-meromorphic map)**.** By definition,

$$X \times_S Y \hookrightarrow X \times Y$$

is an inclusion. Therefore, it is easy to see that

$$S\text{-meromorphic map} \implies \text{meromorphic map}.$$

Conversely, the graph of a meromorphic map $\Gamma \subset X \times Y$ need not be contained in $X \times_S Y$, so that a meromorphic map need not be a S-meromorphic map.

Remark 2.3 (Comparison between S-meromorphic map and fiberwise meromorphic map)**.** Note that a S-bimeromorphic map does not need to be a fiberwise bimeromorphic map, since the restriction of a bimeromorphic map on the subvariety (the fiber) need not be a bimeromorphic map. We will discuss more about the fiberwise bimeromorphic map in the Note-2.

Definition 2.4 (Moishezon space, first definition)**.** A proper, irreducible, reduced analytic space X is Moishezon if it is bimeromorphic to a projective variety X^{p}.

Remark 2.5. The following proposition tells us when the meromorphic map is an actual morphism, using the rigidity lemma.

Let $f : X \dashrightarrow Y$ be a bimeromorphic map with the resolution of indeterminacy.

$$\begin{tikzcd}
 & W \arrow[dl, "p"'] \arrow[dr, "q"] & \\
X \arrow[rr, dashed] & & Y
\end{tikzcd}$$

then if any $C \subset W$ p-exceptional is also q-exceptional. Then the birational map is also a morphism.

Definition 2.6 (Moishezon space, second definition)**.** A proper, irreducible, reduced analytic space X is Moishezon if

$$a(X) := \operatorname{tr}\deg_{\mathbb{C}} M(X) = \dim(X)$$

that is, it has $\dim X$ number of algebraic dependent meromorphic function.

Definition 2.7 (Moishezon space, third definition)**.** A proper irreducible, reduced analytic space X is Moishezon if it carries a big rank 1 reflexive sheaf $\mathscr{F}$. Here the big rank 1 reflexive sheaf means that the induced Kodaira map $g : X \dashrightarrow \mathbb{P}(H^0(X, \mathscr{F}))$ is bimeromorphic onto its image.

Proposition 2.8. The three different definitions for Moishezon spaces above are equivalent.

Proof. see e.g. [Uen75]. □

The first important property for Moishezon space is that it locally looks like a quasi-projective scheme up to an étale cover.

Proposition 2.9 ([Kol22, Proposition 8.2])**.** Let X be a Moishezon space. For every $x \in X$ there is a pointed quasi-projective scheme (x', X') and an étale morphism $(x', X') \to (x, X)$.

Proof. It's quite difficult; for the sake of time, we omit it here. For the curious reader, please refer to [Art70]. □

Lemma 2.10 (Existence of Galois closure)**.** Let $\pi : X' \to X$ be a finite covering between normal analytic varieties. Then there exists a finite Galois covering $\varphi : X'' \to X$ from a normal analytic variety X'' which factors through π which is universal in the following sense:

$$\begin{tikzcd} & X' \arrow[dr, "\pi"] & \\ X'' \arrow[ur] \arrow[rr, "\varphi"'] & & X \end{tikzcd}$$

For any finite Galois covering $\psi : Y \to X$ from a normal analytic variety which factors through π, there exists uniquely a Galois covering $Y \to X''$ over X'.

Using the existence of Galois closure, we can write a normal Moishezon space globally as a quotient of a proper variety by a finite group.

Proposition 2.11 ([Kol22, Proposition 8.3])**.** Let X be a Moishezon variety. If X is normal, then there is a proper variety Y and a finite group G that acts on Y such that $X \cong Y/G$. (Note that in general Y can not be chosen projective.)

Proof. First, by Proposition 2.11, there exists some étale cover of X (indeed, since the étale morphism is finite, we can find an open cover of X by the étale morphism). Since X is proper, we can find some finite cover of it. Now by the previous lemma we can take the Galois closure of the finite étale cover $X_i \to X$. We then apply the universal property of the Galois closure, thus it is possible to patch the collection of Galois closures $\{X_i \to X\}$ together in the Zariski topology via the gluing lemma (see e.g. Hartshorne Exercise II 2.12.), and therefore we can get a finite covering of X, $Y \to X$ and thus $X \simeq Y/G$.

□

Artin [Art70] proved the following theorem, demonstrating the importance of the category of Moishezon spaces in moduli theory.

Proposition 2.12 ([Art70, Theorem 7.3])**.** There is a natural functor

$$\text{an} : (\text{algebraic space of finite type over } \mathbb{C}) \to (\text{complex spaces})$$

extending the functor an on the category (schemes of finite type /$\mathbb{C}$). This functor induces an equivalence of categories

$$\text{(complex algebraic schemes of finite type}/\mathbb{C}) \to \text{(Moishezon spaces)}.$$

In other words, every Moishezon space is in a unique way an algebraic space.

We next prove that a Kahler Moishezon space with 1-rational singularity is a projective variety. Before proving the theorem, let us first state two results that will be used in the proof.

Lemma 2.13. Let X be a compact Moishezon space with 1-rational singularity, that is, X is normal and has a resolution $\pi : Y \to X$ such that $R^1\pi_*\mathcal{O}_Y = 0$. Then an analytic homology class $b \in A_2(X, \mathbb{Q})$ is zero if it is numerically equivalent to 0. In particular,

$$A_2(X, \mathbb{Q}) = N_1(X)_{\mathbb{Q}} \subset H^2(X, \mathbb{Q}).$$

Lemma 2.14 (Nakai-Moishezon criterion for $\mathbb{Q}$-line bundles over Kähler Moishezon space)**.** Let X be a Kähler Moishezon space with a Kähler form ω. Assume that an element $L \in \mathrm{Pic}(X)_{\mathbb{Q}}$ satisfies the equality for any curve $C \subset X$:

$$(C.L) = \int_C \omega.$$

Then L is ample.

Proposition 2.15 ([Nam02])**.** Let X be a Moishezon space with 1-rational singularity. If X is Kähler, then X is projective.

Proof of the Proposition 2.15. Since the numerical equivalence and the homological equivalence coincide for (analytic) 1-cycles by Lemma 2.13, we have a natural map

$$\alpha : N^1(X)_{\mathbb{Q}} \to (A_2(X, \mathbb{Q}))^*, \quad d \mapsto (- \cdot d),$$

and α is an isomorphism (by duality of $N^1(X)_{\mathbb{Q}}$ and $N^1(X)_{\mathbb{Q}}$).

Note that $\omega \in H^2(X, \mathbb{R})$ Kähler form as an element of $(A_2(X, \mathbb{R}))^*$. By simply defining

$$\alpha_\omega : A_2(X, \mathbb{R}) \to \mathbb{R}, \quad C \mapsto \omega \cdot C = \int_C \omega.$$

Since $\alpha_{\mathbb{R}}$ is surjective, there is an element $d \in N^1(X)_{\mathbb{R}}$ such that

$$(C \cdot d) = \int_C \omega,$$

for every curve $C \subset X$.

We then approximate $d \in N^1(X)_{\mathbb{R}}$ by a convergent sequence $\{d_m\}$ of rational elements $d_m \in N^1(X)_{\mathbb{Q}}$.

Let us fix the basis $b_1, \dots, b_l$ of the vector space $N^1(X)_{\mathbb{Q}}$. Each b_i is represented by an element $B_i \in \mathrm{Pic}(X)_{\mathbb{Q}}$ via the quotient

$$\mathrm{Pic}(X)_{\mathbb{Q}} \to N^1(X)_{\mathbb{Q}} = \mathrm{Pic}(X)_{\mathbb{Q}}/\equiv, \quad B_i \mapsto b_i,$$

Now d (resp. d_m) is represented by an element in $\mathrm{Pic}(X)_{\mathbb{R}}$ (resp. $\mathrm{Pic}(X)_{\mathbb{Q}}$)

$$D := \Sigma x_i B_i,$$

(resp. $D_m := \Sigma x_i^{(m)} B_i$) such that $\lim x_i^{(m)} = x_i$. Put $E_m := D_m - D$. Then there are d closed $(1,1)$-forms α_m corresponding to E_m such that $\{\alpha_m\}$ uniformly converge to 0 .

If m is chosen sufficiently large, then $\omega_m := \omega + \alpha_m$ is a Kähler form. Since

$$(C.D_m) = \int_C \omega_m > 0,$$

for every curve $C \subset X$. We see that D_m is ample by Lemma 2.14 (Note that we have D_m being a $\mathbb{Q}$-divisor, so that it's possible to apply the Nakai-Moishezon criterion). In particular, X is projective. □

Remark 2.16. There exist some Kähler Moishezon spaces with bad singularity that are not projective. (As we shall see in the last section).

Proposition 2.17 ([Kol22, Proposition 8])**.**
(1) Let X be a Moishezon space, if $Z \to X$ be finite then Z is Moishezon.

(2) Let X be a Moishezon space, and $f : X \to Y$ be a surjective morphism of complex varieties. Then Y is also Moishezon.

(3) Let X be a Moishezon space, assume that $Z \subset X$ is Moishezon, then

Proof of (1). By definition

$$\mathrm{trdeg}_{\mathbb{C}} K(X) = \dim X,$$

and if Z is finite map then

$$K(X) \hookrightarrow K(Z),$$

is a finite field extension. Therefore by additive property for a tower of field extensions, we have

$$\mathrm{trdeg}_{\mathbb{C}}(K(Z)) = \mathrm{trdeg}_{\mathbb{C}}(K(X)) + \mathrm{trdeg}_{K(X)} K(Z) = \mathrm{trdeg}_{\mathbb{C}}(K(X)).$$

□

Proof of (2). It will be generalized in to the relative version, see 3.14. □

Proof of (3). Consider the following pull back diagram.

$$\begin{tikzcd} Z^p = f^{-1}(Z) \arrow[r] \arrow[d, "f_Z"'] & X^p \arrow[d, "f"] \\ Z \arrow[r, hook] & X \end{tikzcd}$$

Clearly Z^p is projective (as subvariety of X^p), and f_Z is surjective (by definition of Z^p). Therefore, by (2), we know that Z is again Moishezon. □

The following proposition shows that the Moishezon manifolds admit strong Hodge decomposition.

Proposition 2.18. If X is a Moishezon manifold, then the Hodge decomposition holds, indeed a Moishezon manifold admits strong Hodge decomposition.

Before proving the theorem, let us first define what a strong Hodge decomposition is. We say that a compact manifold admits a *strong Hodge decomposition* if the natural maps

$$H^{p,q}_{\mathrm{BC}}(X,\mathbb{C}) \longrightarrow H^{p,q}(X,\mathbb{C}),\ [\alpha^{p,q}]_{\mathrm{BC}} \mapsto [\alpha^{p,q}]_{\bar\partial} \quad \bigoplus_{p+q=k} H^{p,q}_{\mathrm{BC}}(X,\mathbb{C}) \longrightarrow H^k(X,\mathbb{C}), \quad \sum[\alpha^{p,q}] \mapsto \sum \alpha^{p,q},$$

are isomorphisms.

Remark 2.19. As a direct consequence, we see that a Moishezon manifold admits *the Du Bois property*, that is
$$H^i(X,\mathbb{C}) \to H^i(X,\mathcal{O}_X),$$
is surjective for all $i \geq 0$. (which will be used in the third note).

Proof. The idea of the proof comes from [Dem97, Proposition (12.3)]. We first take the projective modification
$$\mu : \widetilde{X} \to X,$$
such that X' is a projective manifold. And therefore X' admits a strong Hodge decomposition. On the other hand

We first observe that $\mu_\star\mu^\star\beta = \beta$ for every smooth form β on Y. In fact, this property is equivalent to the equality
$$\int_Y (\mu_\star\mu^\star\beta)\wedge\alpha = \int_X \mu^\star(\beta\wedge\alpha) = \int_Y \beta\wedge\alpha.$$
for every smooth form α on Y, and this equality is clear because μ is a biholomorphism outside sets of Lebesgue measure 0 (which holds in general for a proper surjective bimeromorphic map).

Consequently, the induced cohomology morphism $\mu_\star$ is surjective and $\mu^\star$ is injective (but these maps need not be isomorphisms).

$$\begin{array}{ccccccc}
H^{p,q}_{\mathrm{BC}}(\widetilde{X},\mathbb{C}) & \longrightarrow & H^{p,q}(\widetilde{X},\mathbb{C}), & \quad & \bigoplus\limits_{p+q=k} H^{p,q}_{\mathrm{BC}}(\widetilde{X},\mathbb{C}) & \longrightarrow & H^k(\widetilde{X},\mathbb{C}) \\
\mu_\star\downarrow\uparrow\mu^\star & & \mu_\star\downarrow\uparrow\mu^\star & & \mu_\star\downarrow\uparrow\mu^\star & & \mu_\star\downarrow\uparrow\mu^\star \\
H^{p,q}_{\mathrm{BC}}(X,\mathbb{C}) & \longrightarrow & H^{p,q}(X,\mathbb{C}), & & \bigoplus\limits_{p+q=k} H^{p,q}_{\mathrm{BC}}(X,\mathbb{C}) & \longrightarrow & H^k(X,\mathbb{C})
\end{array}$$

Now, we have commutative diagrams with either upward or downward vertical arrows. Hence the surjectivity or injectivity of the top horizontal arrows implies that of the bottom horizontal arrows.

□

We next introduce Campana's Moishezon criterion. The proof uses the core reduction he introduced.

Proposition 2.20 ([Cam81, Corollaire on p. 212])**.** Let X be a compact complex variety in the Fujiki class $\mathscr{C}$. Then X is Moishezon if and only if X is algebraically connected.

As an immediate consequence.

Corollary 2.21. A compact Kahler manifold is projective iff it's algebraically connected.

Proposition 2.22. Let $f : X \to B$ be a fibration over an algebraically connected variety (e.g. a projective curve). Assume that X is in the Fujiki class $\mathscr{C}$ and the general fiber of f is algebraically connected, then X is Moishezon if and only if f has a multi-section.

Proof. The proof is clear, since admitting a multi-section implies the algebraic connectedness of X. □

Remark 2.23. For readers interested in the applications of the algebraic connectedness criterion, I recommend the paper by [Lin23]. He tries to address the following question.

Question 2.24 (Oguiso–Peternell problem, [Lin23, Problem 1.2])**.** Let X be a compact Kähler manifold of dimension n such that $\operatorname{Int}(\operatorname{Psef}(X)^\vee)$ (or $\operatorname{Int}(\mathscr{K}(X)^\vee)$ for dual Kähler cone $\mathscr{K}(X)$) contains an element of $H^{2n-2}(X, \mathbb{Q})$. Is X always projective? If not, how algebraic is X ?

3 Moishezon morphisms

Let us first recall the definition of a projective morphism.

Definition 3.1 (Projective morphism, first definition)**.** Let $X \to S$ be a proper morphism between complex spaces. f is projective if there exists a locally free coherent sheaf $\mathcal{E}$ of finite rank such that there exists a closed S-immersion $X \hookrightarrow \mathbb{P}_S(\mathcal{E})$, with the following diagram commuting.

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & \mathbb{P}_S(\mathcal{E}) \arrow[dl] \\ & S & \end{tikzcd}$$

Definition 3.2 (Projective morphism, second definition)**.** Let $X \to S$ be a proper morphism between complex spaces. f is projective if X can be embedded in $\mathbb{P}^N \times S$ for some N, with the following diagram commuting.

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & \mathbb{P}^N \times S \arrow[dl] \\ & S & \end{tikzcd}$$

Note that Kollár adopts the second definition.

Definition 3.3 (Locally projective morphism)**.** Let $f : X \to S$ be a proper morphism of complex spaces. We call f locally projective if for every relatively compact open subset Q of S the restriction $f_Q : X_Q \to Q$ is a projective morphism.

Remark 3.4. It is easy to see the second definition immediately implies the first definition. The converse direction also holds when the base is Stein or quasi-projective.

Proof. Assume we have the 1st definition, so that $f : X \to S$ and $g : Y = \mathbb{P}_S(f_*\mathcal{L}^{\otimes m}) \to S$. Let A be an g-ample line bundle. And, therefore by Serre vanishing theorem over some Stein compact subset $B \subset S$, for some sufficiently large $n \gg 0$, we have

$$g^*g_*(\mathcal{E} \otimes A^{\otimes n}) \to \mathcal{E} \otimes A^{\otimes n},$$

is surjective. Since the base S is Stein, by Cartan A theorem, $g_*(\mathcal{E} \otimes A^{\otimes n})$ is globally generated. And therefore so is the pull back $g^*g_*(\mathcal{E} \otimes A^{\otimes n})$. Since the surjective map sends a globally generated coherent sheaf to a globally generated coherent sheaf, this means that $\mathcal{E} \otimes A^{\otimes n}$ is globally generated.

By coherence of $\mathcal{E} \otimes A^{\otimes n}$, the cohomology group $V = H^0(Y, \mathcal{E} \otimes A^{\otimes n})$ is finite dimensional. And there is a surjection

$$V \otimes \mathcal{O}_Y \to \mathcal{E} \otimes A^{\otimes n}.$$

And therefore it will induce an embedding

$$X \hookrightarrow \mathbb{P}_B(\mathcal{E}) = \mathbb{P}_B\left(\mathcal{E} \otimes A^{\otimes m}\right) \hookrightarrow \mathbb{P}(V) \times B,$$

after shrinking the base $B \subset S$. $\square$

Remark 3.5. When the total space has only a finite number of irreducible components, then a locally projective morphism is bimeromorphic to a projective morphism. (see [Fuj83, Lemma 1.3.1]).

In what follows, we may assume that the base S is reduced. However, in general, we do not require the total space X to be reduced or not.

Definition 3.6 (Moishezon morphism, 1st definition)**.** A proper morphism of analytic spaces $g : X \to S$ is Moishezon if $g : X \to S$ is bimeromorphic to a projective morphism $g^{\mathrm{p}} : X^{\mathrm{p}} \to S$.

That is, there is a closed subspace $Y \subset X \times_S X^{\mathrm{p}}$ such that the coordinate projections $Y \to X$ and $Y \to X^{\mathrm{p}}$ are bimeromorphic.

$$\begin{tikzcd}
 & Y \arrow[dl] \arrow[dr] & \\
X \arrow[rr, dashed] \arrow[dr] & & X^p \arrow[dl] \\
 & S &
\end{tikzcd}$$

Definition 3.7 (Moishezon morphism, 2nd definition)**.** A proper morphism of analytic spaces $g : X \to S$ is Moishezon if there is a projective morphism of algebraic varieties $G : \mathbb{X} \to \mathbb{S}$ and a meromorphic $\phi_S : S \dashrightarrow \mathbb{S}$ such that X is bimeromorphic to $\mathbb{X} \times_{\mathbb{S}} S$, the fiber product of rational maps is defined where the maps are defined, so on a dense open set.

Remark 3.8. Let us say a few words about the fiber product for a rational map $\phi_S : S \dashrightarrow \mathbb{S}$, the fiber product is defined on the place that ϕ_S is a holomorphic map.

Definition 3.9 (Moishezon morphism, 3rd definition)**.** A proper morphism of analytic spaces $g : X \to S$ is Moishezon if there is a rank 1, reflexive sheaf L on X such that the natural map $X \dashrightarrow \operatorname{Proj}_S(g_*L)$ is bimeromorphic onto the closure of its image.

Proposition 3.10. The three definitions of Moishezon morphism are equivalent.

Proof. Definition 3.7 equivalent to Definition 3.6 is clear (using Proposition 3.16). Conversely, if there exists a projective family $X^p \to S$ that is bimeromorphic to a given $f : X \to S$, then by generic flatness we know $g^p : X^p \to S$ is flat over S^o for some Zariski open subset $S^o \subset S$, and therefore using the definition of projective family, there exists a morphism

$$S^o \to \mathrm{Hilb}(\mathbb{P}^N)$$

such that the projective family is the pull back

$$\begin{tikzcd} X^p \arrow[r] \arrow[d] & \mathcal{M} \arrow[d] \\ S \arrow[r, dashed] & \mathrm{Hilb}(\mathbb{P}^N) \end{tikzcd}$$

We now show that the first definition and third definition are equivalent. From the third definition to the first definition is clear since $\mathrm{Proj}_S(f_*L)$ is projective over S. Conversely, if $f : X \to S$ is bimeromorphic to a projective morphism $X^p \to S$. Then since we assume X is normal, therefore the meromorphic map $X \dashrightarrow X^p$ is a morphism outside a codimension 2 subset. And the pull back $(\phi^o)^*\mathcal{O}_X(1)$ is a big line bundle defined on a big open subset, which can be extended uniquely to a big rank 1 reflexive sheaf. $\square$

Remark 3.11. The terminology in different papers is different; we can summarize it as below.

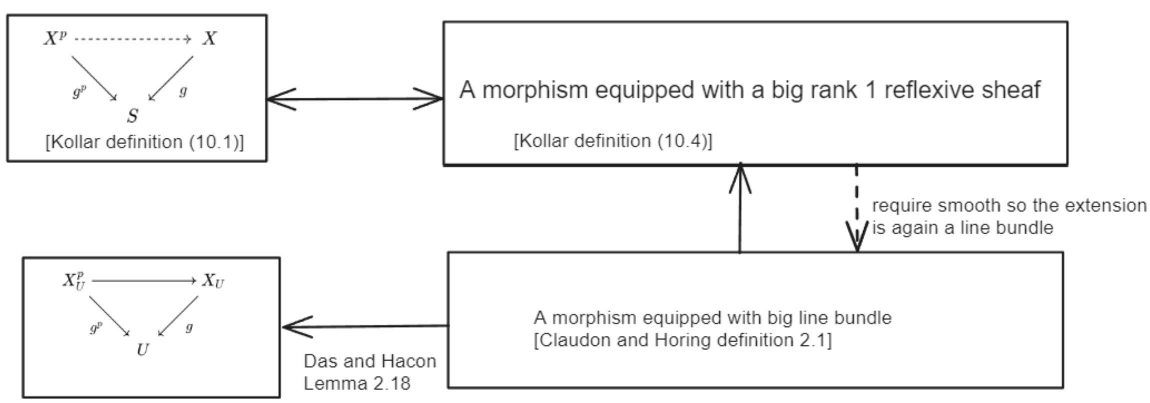

Figure 1: Definitions in different papers

The Moishezon morphism satisfies the following Chow type lemma (which can be viewed as the deterministic property of a Moishezon morphism).

Theorem 3.12 ([DH20, Lemma 2.18])**.** Let $f : X \to S$ be a proper surjective morphism of analytic varieties, and let L be a f-big line bundle on X and D a $\mathbb{Q}$-divisor. Then

(1) Over any relatively compact open subset $V \subset S$, there exists a proper (indeed it's projective see [CH24]) bimeromorphic morphism $\alpha : W \to f^{-1}V$ from a smooth analytic variety W such that

$\beta = f|_{f^{-1}V} \circ \alpha : W \to V$ is a projective morphism and,
(2) $\left(W, \alpha_*^{-1}\left(D|_{f^{-1}V}\right) + \mathrm{Ex}(\alpha)\right)$ is a log smooth pair.

First, let us compare the theorem above with Definition 3.6; in the Definition, we only assume the existence of some bimeromorphic S-map, whereas the Chow lemma allows us to choose some bimeromorphic projective morphism.

Proof. Let $\phi : X \dashrightarrow Y$ be the relative Iitaka fibration of L over S and $g : Y \to S$ the induced projective morphism. Since L is f-big, $\phi : X \dashrightarrow Y$ is bimeromorphic. Let $p : \Gamma \to X$ and $q : \Gamma \to Y$ be the resolution of indeterminacy of ϕ so that p is proper.

$$\begin{tikzcd} & \Gamma \arrow[dl, "p"'] \arrow[dr, "q"] & \\ X \arrow[rr, dashed, "\phi"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

Now fix a relatively compact open subset $V \subset S$. Choose another relatively compact open set $U \subset S$ containing V such that $\bar{V} \subset U$. Note that U is σ-compact, since it is relatively compact. Since f and g are both proper morphisms, it follows that $X_U := f^{-1}U$ and $Y_U := g^{-1}U$ are both σ-compact. Let $\Gamma_U := q^{-1}\left(g^{-1}U\right) = p^{-1}\left(f^{-1}U\right)$. Then from the commutative diagram above it follows that $q|_{\Gamma_U} : \Gamma_U \to g^{-1}U$ is a proper morphism. In particular, Γ_U is σ-compact. Note that $q|_{\Gamma_U}$ is bimeromorphic. Therefore there is a projective bimeromorphic morphism $h : Z \to \Gamma_U$ from an analytic variety Z such that $q|_{\Gamma_U} \circ h : Z \to Y_U$ is a projective bimeromorphic morphism. Since g is projective, so is $Z \to U$.

Now we replace U by our previously fixed open set V. Then $Z_V := (g \circ\, q \circ h)^{-1}V$ is a relatively compact open subset of Z. Let $r : W \to Z_V$ be the log resolution of $\left(Z_V, (p \circ h)_*^{-1}\left(D|_{f^{-1}V}\right)\right)$. Let $\alpha := p|_{\Gamma_V} \circ h\Big|_{h^{-1}\Gamma_V} \circ r$ and $\beta := g|_{g^{-1}V} \circ q\Big|_{\Gamma_V} \circ h\Big|_{h^{-1}\Gamma_V} \circ r$, where $\Gamma_V := p^{-1}\left(f^{-1}V\right) = q^{-1}\left(g^{-1}V\right)$. Note that β is a projective morphism, since it is a composition of projective morphisms over relatively compact bases.

Then $\alpha : W \to f^{-1}V$ is a proper bimeromorphic morphism and $\beta : W \to V$ is a projective morphism such that $\beta = f|_{f^{-1}V} \circ \alpha$ and $\left(W, \alpha_*^{-1}\left(D|_{f^{-1}V}\right) + \mathrm{Ex}(\alpha)\right)$ is a log smooth pair. $\square$

Proposition 3.13 ([Fuj83, Proposition 1.5.(4)])**.** Suppose that there exists a locally projective morphism $g : Y \to S$ and a generically finite meromorphic S-map $h : X \dashrightarrow Y$. Then f is Moishezon.

$$\begin{tikzcd} X \arrow[rr, dashed, "h"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

Proof. First, since being Moishezon is stable under bimeromorphic change, without loss of generality we can assume that h is a morphism. Since Moishezon morphism and locally projective morphism are proper, h is proper. Apply the Stein factorization theorem, such that h_2 is projective (since h_2 is finite) and h_1 is proper. Thus, the composition $g \circ h_2$ is locally projective. And thus by definition $X \to S$ is a Moishezon morphism.

$$\begin{tikzcd} & X^* \arrow[dr, "h_2"] & \\ X \arrow[ur, "h_1"] \arrow[rr, "h"] \arrow[dr] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

□

Proposition 3.14 ([Fuj83, Proposition 1.7])**.** Let $f : X \to S$ be a Moishezon morphism, and $g : Y \to S$ a proper morphism, of reduced complex spaces. Suppose that there is a generically surjective meromorphic S-map $h : X \dashrightarrow Y$. Then g also is Moishezon.

Proof. This Proposition can be viewed as a generalization of Proposition 2.17. The proof is a bit involved, and we omit it here. □

Proposition 3.15 ([Fuj83, Proposition 1.5])**.**
(1) The morphism $f : X \to S$ is Moishezon if and only if for each irreducible component X_i of X the restriction $f = f|_{X_i} : X_i \to S$ is Moishezon.

(2) Let $f : X \to S$ be a Moishezon morphism. Then: For every reduced analytic subspace $X' \subseteq X$ the induced morphism $f' = f|_{X'} : X' \to S$ is Moishezon.

Proof. For (1), let's take the normalization

$$\nu : X^\nu \to X,$$

recall that for a reduced complex space with finitely many irreducible components, the normalization is a bimeromorphic map. So that $f : X \to S$ is Moishezon iff the restrictions on each component X_i are Moishezon.

For (2), by the Chow lemma (Theorem 3.12), we can find some locally projective morphism such that X^* is smooth and h is a bimeromorphic S-morphism.

$$\begin{tikzcd} X^* \arrow[rr, "h"] \arrow[dr, "g"'] & & X \arrow[dl, "f"] \\ & S & \end{tikzcd}$$

We then take the inverse image of the analytic subspace X' and denote it $Z = h^{-1}(X')$. (we can assume the inverse image has reduced structure). Since the restriction of the projective morphism on $g|_Z : Z \to S$ is still locally projective. And by construction, clearly the morphism $Z \to X'$ is surjective. And therefore, by Proposition 3.14, we know that $X' \to S$ is a Moishezon morphism. □

Restriction on the image side will also preserve the Moishezon condition.

Proposition 3.16 (A morphism is Moishezon iff it is Moishezon onto its image)**.** Let $f : X \to S$ be a proper morphism between analytic spaces, let $f' : X \to f(X) = Y \subset S$ be the restriction, then f is Moishezon (resp. projective) iff f' is Moishezon (resp. projective).

Proof. It's enough to prove the case for the projective morphism case (and the Moishezon morphism case follows easily).

To see this, assume that the morphism $f : X \to S$ is projective, by definition it factors through $X \hookrightarrow \mathbb{P}^n_S \to S$. Doing base change on $Y \hookrightarrow S$, proves the projectivity of $X \hookrightarrow \mathbb{P}_Y \to Y$.

The converse direction is clear, since the composition $X \to f(X) \hookrightarrow S$ can be written as:

$$X \hookrightarrow \mathbb{P}^n_{f(X)} \to f(X) \hookrightarrow S.$$

Since $\mathbb{P}^n_{f(X)} = \mathbb{P}^n_S \times_S f(X)$, we can rewrite the morphism as:

$$X \hookrightarrow \mathbb{P}^n_S \times_S f(X) \hookrightarrow \mathbb{P}^n_S \to S,$$

where the second inclusion is because $f(X) \hookrightarrow S$ and so is the projective bundle. $\square$

Proposition 3.17. When the base is Moishezon, then the total space is Moishezon iff the morphism is Moishezon.

Proof. We first prove that a morphism between Moishezon spaces is a Moishezon morphism. Let us define the graph embedding to be

$$\iota : X \to X \times S, \quad x \mapsto (x, f(x)),$$

since X is Moishezon it's bimeromorphic to a projective variety, as the diagram below shows

$$\begin{tikzcd} X \arrow[r, hook, "\iota"] \arrow[dr, "f"'] & X \times S \arrow[r, dashed] \arrow[d, "\pi"] & X^p \times S \arrow[dl, "\pi^p"] \\ & S & \end{tikzcd}$$

Clearly, π^p is a projective morphism. And consequently π is a Moishezon morphism. And finally by Proposition 3.15, the morphism $f : X \to S$ is again Moishezon.

Conversely, if the morphism is Moishezon, and S is a Moishezon space, then there exist bimeromorphic modifications such that the following diagram commutes

$$\begin{tikzcd} X^p \arrow[r] \arrow[d] & X' \arrow[r] \arrow[d] & X \arrow[dl] \\ S^p \arrow[r] & S & \end{tikzcd}$$

Where $X' \to S$ is a projective morphism and S^p is a projective variety. Since the base change preserves the projective condition, it is easy to see that $X^p \to S^p$ is a projective morphism over S^p. And therefore X^p is a projective variety. By Proposition 3.14, X' is a Moishezon space. Since $X' \to X$ is bimeromorphic, this implies that X is also Moishezon. $\square$

Proposition 3.18 ([Kol22, Lemma 15])**.** Let $g : X \to S$ be a proper, generically finite, dominant morphism of normal, complex, analytic spaces. Then $\mathrm{Ex}(g) \to S$ is Moishezon.

Proof. We will prove the result under the additional assumption that S is Stein. By the geometric Noether normalization theorem, there exists a finite morphism

$$S \to \mathbb{C}^{\dim S}.$$

After replacing the base by $\mathbb{C}^{\dim S}$, we can assume that the smooth locus of S is dense in $g(\mathrm{Ex}(g))$. Note that, by Proposition 3.13, if the restriction on $\mathbb{C}^{\dim S}$ is Moishezon, then so will the restriction on S. We will prove the result by induction on dimension.

We first define the base case $(g_0 : X_0 \to S_0) := (g : X \to S)$. Let E_0 be a g_0 exceptional divisor, with the image $Z_0 = g_0(E_0)$. We then inductively define the morphism $g_{i+1} : X_{i+1} \to S_{i+1}$ as follows. Assume that we already constructed $g_i : X_i \to S_i$; we then blow up S_i along Z_i and let S_{i+1} be the normalization of the blow-up $\mathrm{Bl}_{Z_i} S_i$. Since S_i is reduced, this will induce a generically finite map $\phi : X_i \dashrightarrow S_{i+1}$. So that by the universal property of the normalization, the generically finite morphism $g_i : X_i \to S_i$ lifts to a generically finite morphism $g_{i+1} : X_{i+1} \to S_{i+1}$, where X_{i+1} is the normalization of the graph of the map $\phi : X_i \dashrightarrow S_{i+1}$.

$$\begin{tikzcd}
X_{i+1} \arrow[rr, "g_{i+1}"] \arrow[d] & & S_{i+1} = \mathrm{Bl}_{Z_i}(S_i)^\nu \arrow[d] \\
\Gamma_\phi \arrow[urr] \arrow[d] & & \mathrm{Bl}_{Z_i}(S_i) \arrow[d] \\
X_i \arrow[rr, "g_i"'] \arrow[uurr, dashed, "\phi"] & & S_i
\end{tikzcd}$$

Let $E_{i+1} \subset X_{i+1}$ denote the bimeromorphic transform of E_i. (Note that $X_{i+1} \to X_i$ is an isomorphism over an open subset of E_i). We then compute the vanishing order $a(E_i, S_i)$ of the Jacobian of g_i along E_i. We claim that

$$a\left(E_{i+1}, S_{i+1}\right) \leq a\left(E_i, S_i\right) + 1 - \operatorname{codim}\left(Z_i \subset S_i\right).$$

Thus eventually we reach the situation when $\operatorname{codim}(Z_i \subset S_i) = 1$, indeed if $\operatorname{codim}(Z_i \subset S_i) \geq 2$ then the Jacobian of g_i along E_i will eventually go to zero. Contradiction.

Thus by comparing the dimension we know when we restrict the morphism $X_i \to S_i$ to $E_i \to Z_i$ it will become a generically finite morphism. Since $S_{i+1} \to S_i$ is projective, the composition $Z_i \to Z_0$ will be a locally projective morphism.

$$\begin{tikzcd}
E_i \arrow[rr] \arrow[dr] & & Z_i \arrow[dl] \\
& Z_0 &
\end{tikzcd}$$

By Proposition 3.13, we know that $E_i \to Z_0$ is a Moishezon morphism. Since the strict transform $E_i \to E_0$ is a dominant morphism, by Proposition 3.14, we know that $E_0 \to Z_0$ is also a Moishezon morphism. Finally, by Proposition 3.15 and Proposition 3.16, we know that $\mathrm{Ex}(f) \to S$ is Moishezon. ☐

Theorem 3.19 (Fibers of the Moishezon morphism are Moishezon spaces, [Kol22, Corollary 16])**.** The fibers of a proper, Moishezon morphism are Moishezon.

Proof. Let $g : X \to S$ be a proper, Moishezon morphism. It is bimeromorphic to a projective morphism $X^{\mathrm{p}} \to S$. We may assume X^{p} to be normal. Let Y be the normalization of the closure of the graph of $X \dashrightarrow X^{\mathrm{p}}$.

$$\begin{tikzcd} & Y \arrow[dl] \arrow[dr] & \\ X \arrow[rr, dashed] \arrow[dr] & & X^p \arrow[dl] \\ & S & \end{tikzcd}$$

Fix now $s \in S$. Let $Z_s \subset X_s$ be an irreducible component; since given a proper dominant morphism, there exists at least one irreducible component dominating the base, there exists $W_s \subset Y_s$ an irreducible component that dominates Z_s. And by Proposition 3.14 and Proposition 3.15, it's enough to show that W_s is Moishezon. We divide the problem into two cases:

If $\pi : Y \to X^{\mathrm{p}}$ is generically an isomorphism along W_s, then W_s is bimeromorphic to an irreducible component of X^{p}_s, hence Moishezon.

Otherwise $W_s \subset \mathrm{Ex}(\pi)$. Now $\mathrm{Ex}(\pi) \to X^{\mathrm{p}}$ is Moishezon by Proposition 3.18. And by induction on dimension, since $\dim \mathrm{Ex}(\pi) < \dim X = \dim Y$, the fiber W_s is Moishezon. ☐

Proposition 3.20 ([Kol22, Example 13])**.** Let Z be a normal, projective variety with discrete automorphism group. Let $g : X \to S$ be a fiber bundle with fiber Z over a connected base S. Then g is Moishezon $\Leftrightarrow$ g is projective $\Leftrightarrow$ the monodromy is finite.

Remark 3.21. The monodromy here is different from the cohomological monodromy. Here the monodromy is referred to as the fiber bundle monodromy

$$\rho : \pi_1(S) \to G \subset \mathrm{Aut}(Z)$$

where $G \subset \mathrm{Aut}(Z)$ is the structure group of the fiber (e.g. when the fiber bundle is a principal G-bundle, then the structure group is simply the group G). Finite monodromy condition means that $\mathrm{im}(\rho) \subset G$ is a finite subgroup.

Before proving the theorem, let us state a lemma from fiber bundle theory that is useful in what follows (which can be viewed as a generalization of the Ehresmann theorem over a simply connected base: when the base is simply connected, the fiber bundle is automatically trivial).

Lemma 3.22. Let $g : X \to S$ be a fiber bundle with trivial monodromy group, then the fiber bundle is actually a trivial fiber bundle.

The following proposition about the automorphism of a polarized projective variety will be useful in the proof.

Proposition 3.23. Let Z be a projective variety, and L an ample line bundle over Z. Then $\mathrm{Aut}(Z, L) = \{\phi \in \mathrm{Aut}(Z) \mid c_1(\phi^* L) = c_1(L)\}$ is finite if the automorphism group $\mathrm{Aut}(Z)$ is discrete.

Proof. Let
$$\phi : Z \to Z \in \mathrm{Aut}(Z, L),$$
consider the graph $\Gamma_\phi \subset Z \times Z$, thus the Hilbert polynomial of Γ_ϕ relative to $L \boxtimes L$ is given by
$$H_\phi(n) = \chi(\Gamma_\phi, (L \boxtimes L)^n) = \chi(Z, L^{\otimes n} \otimes \phi^*(L^{\otimes n})).$$
On the other hand, since $c_1(\phi^* L) = c_1(L)$, it is easy to see that $L^{\otimes n} \otimes \phi^*(L^{\otimes -n})$ is numerically trivial, and thus
$$H(n) = \chi(Z, L^{\otimes n} \otimes \phi^*(L^{\otimes n}) \otimes L^{\otimes n} \otimes \phi^*(L^{\otimes -n})) = \chi(Z, L^{\otimes 2n}),$$
which is independent of ϕ and denote it $P(n) = H_\phi(n)$. So that the graph lies on $\mathrm{Hilb}^P_{Z \times Z}$ (with fixed Hilbert polynomial $P(n)$), which is of finite type. Thus it contains finitely many irreducible components (by the Noetherian property). On the other hand, since Z is projective, each irreducible component of the Hilbert scheme is proper. Thus $\mathrm{Aut}(Z, L)$ is finite. □

The idea of the proof of the theorem is provided by Professor Kollár.

Proof of the theorem. We only need to show that (1) implies (3) and (3) implies (2). For (3) implies (2), we try to take an étale base change so that the fiber bundle becomes a trivial bundle. We can do this as follows. Consider $\rho(\pi_1(S)) \subset \mathrm{Aut}(Z)$ is finite, let $S' \to S$ be the corresponding finite (unbranched) cover that kills the monodromy. Indeed since we have
$$\rho : \pi_1(S) \to G$$
then the kernel $\ker(\rho)$ is a subgroup of $\pi_1(S)$ of finite index, therefore by the Galois correspondence for coverings, there exists a finite étale cover of the base
$$\tilde{S} \to S,$$
such that the monodromy of the fiber bundle under the base change becomes trivial, then by the previous lemma, after the base change the fiber bundle becomes a trivial bundle
$$Z \times \tilde{S} \to \tilde{S},$$
clearly the morphism is projective and admits a relative ample line bundle L (since Z is projective). And therefore if we define
$$L' = \bigotimes_{g \in \Gamma} g^* L.$$
Since L' is monodromy invariant, the ample line bundle will descend to the original fiber bundle $g : X \to S$ and thus g is a projective morphism.

For (1) implies (3), since $g : X \to S$ is Moishezon, by Definition 3.9, there exists a g-big (rank 1 reflexive sheaf) H on X (since it's a fiber bundle, the restriction on Z is again big, and we denote it also as H). Given an ample line bundle L on the fiber Z, we consider the monodromy action on L, which pulls back the ample line bundle to another ample line bundle $L_\gamma = \rho(\gamma)^* L$.

Note that under the monodromy action, the intersection

$$d := H \cdot (L_\gamma)^{n-1},$$

remains the same for all γ.

We then consider the linear functional

$$\ell : N^1(Z)_{\mathbb{R}} \to \mathbb{R}, \quad M \mapsto M^{n-1} \cdot H,$$

if we restrict the linear functional on the ample cone $\mathrm{Amp}(Z)$, then

$$S_d = \{M \in N^1(Z)_{\mathbb{R}} \cap \mathrm{Amp}(Z) \mid M^{n-1} \cdot H = d\},$$

is a bounded slice. To see this, by the Khovanskii-Teissier inequality we have

$$\left(H \cdot M^{n-1}\right)^n \geq (H^n)\,(M^n)^{n-1},$$

thus we get

$$\mathrm{vol}(M) \leq \left(\frac{d^n}{H^n}\right)^{\frac{1}{n-1}},$$

so that it is bounded (for a fixed H). Thus the slice contains only finitely many lattice points of $\mathrm{NS}(Z)$.

$$\#\{M \in \mathrm{NS}(Z) \cap \mathrm{Amp}(Z) \mid \ell(M) = d\} < \infty.$$

In particular, the ample line bundle on the monodromy orbit is finite

$$\Gamma \cdot L = \{L_\gamma := \rho(\gamma) \cdot L \mid \gamma \in \pi_1(S)\} \subset \{M \in \mathrm{NS}(Z) \cap \mathrm{Amp}(Z) \mid \ell(M) = d\}.$$

This will force the monodromy to be finite, indeed apply the orbit-stabilizer theorem

$$\boxed{|\Gamma| = |\Gamma \cdot L||\mathrm{Stab}(L)| < +\infty}$$

Thus we only need to prove that $\mathrm{Stab}(L) = \mathrm{Aut}(Z, L) = \{\phi \in \mathrm{Aut}(Z) \mid \phi^* L = L\}$ is finite. On the other hand, since $\mathrm{Aut}(Z)$ is discrete, by Proposition 3.23, this means that $\mathrm{Stab}(L) = \mathrm{Aut}(Z, L)$ is finite. □

4 Examples

In this section, we will present various examples related to the Moishezon space and Moishezon morphism.

4.1 The Hironaka's example

Hironaka discovered a bunch of complete non-projective 3-folds, which are called Hironaka's varieties. Note that based on the construction of Hironaka, we can from almost all the projective varieties construct some Moishezon spaces, which is why we said at the beginning that Moishezon spaces are versatile in birational geometry. (However, this is not true in dimension 2, since all the smooth Moishezon surfaces are actually projective, see e.g. [GPR94]). The major reference of this part of the note is the paper by Ulrich Thiel (see `https://ulthiel.com/math/wp-content/uploads/other/hironakas_example.pdf`).

Given a smooth projective threefold, which contains two rational curves transversely intersecting at two points. Assume that the two rational curves are C and D that intersect at the points P, Q.

We then take two steps, blow up

$$\begin{aligned} X_1 &= \mathrm{Bl}_{(D\backslash P)'}\left(\mathrm{Bl}_{C\backslash P}(X\backslash P)\right) \xrightarrow{\pi_2} \mathrm{Bl}_{C\backslash P}(X\backslash P) \xrightarrow{\pi_1} X\backslash P \\ X_2 &= \mathrm{Bl}_{(C\backslash Q)'}\left(\mathrm{Bl}_{D\backslash Q}(X\backslash Q)\right) \xrightarrow{\sigma_2} \mathrm{Bl}_{D\backslash Q}(X\backslash Q) \xrightarrow{\sigma_1} X\backslash Q, \end{aligned}$$

Note that if we define $U = X - \{P, Q\}$, then $\pi^{-1}(U) \cong \sigma^{-1}(U)$. In particular, we can glue X_1 and X_2 along $\pi^{-1}(U)$ and $\sigma^{-1}(U)$. In the picture below, we glue the red exceptional surface on the right hand side with the black exceptional surface on the left hand side (denote it S_1) and the blue exceptional surface on the left hand side with the black exceptional surface on the right hand side (denote it S_2). (see picture 2). By the gluing lemma, there exists a morphism $f : H \to X$ and the restriction of the morphism on S_1, S_2 as $f_1 = f|_{S_1} : S_1 \to C$ and $f_2 = f|_{S_2} : S_2 \to C$.

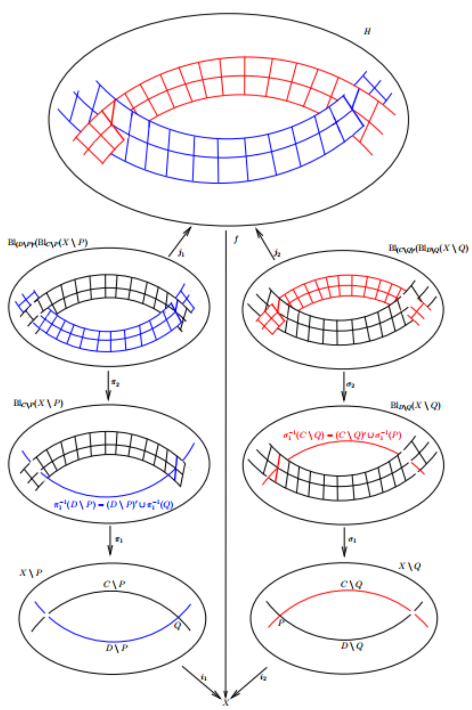

Figure 2: Construction of the Hironaka's variety

We claim that Hironaka's variety is non-projective. The idea to prove the non-projectivity is to find some curve on the surface $S = S_1 \cup S_2$ which has positive degree but adds up to 0.

The key observation is that $f^{-1}(P)$ (resp. $f^{-1}(Q)$) decomposes into two split projective lines L_Q and L'_Q in S_1 (resp. L_P and L'_P in S_2). (see the precise statement below).

Choose two points $A \in C - \{P, Q\}$ and $B \in D - \{P, Q\}$. Since all the points on a rational curve are linearly equivalent, therefore

$$\begin{aligned} A \sim_C Q &\Longrightarrow f_1^{-1}(A) \sim_{S_1} f_1^{-1}(Q) = L_Q + L'_Q \\ B \sim_D P &\Longrightarrow f_2^{-1}(B) \sim_{S_2} f_2^{-1}(P) = L_P + L'_P \end{aligned}$$

and pushing forward the cycle, we get equivalence on S.

$$\begin{aligned} I : f^{-1}(A) &\sim_S f^{-1}(Q) = L_Q + L'_Q \\ II : f^{-1}(B) &\sim_S f^{-1}(P) = L_P + L'_P \end{aligned}$$

On the other hand we also have that B, Q lie on the same rational curve, so that

$$III : B \sim_D Q \Rightarrow f_2^{-1}(B) \sim_{S_2} f_2^{-1}(Q) \Rightarrow f^{-1}(B) \sim_S L'_Q$$

and combining them together, we get

$$\begin{aligned} f^{-1}(A) + f^{-1}(B) \sim_S f^{-1}(A) + f^{-1}(B) &\Rightarrow L_Q + L'_Q + L_P + L'_P \sim_S L'_Q + L'_P \\ &\Rightarrow L_Q + L_P \sim_S 0 \end{aligned}$$

If there exists some ample divisor on A, then both $L_Q \cdot A > 0$ and $L_P \cdot A > 0$ contradict the linearly trivial relation above. Therefore the only possible case is that Hironaka's variety is non-projective.

4.2 Flop the lines on general quintic threefold produce Moishezon variety

4.3 Locally Moishezon morphism which is not globally Moishezon

There are rational and K3 surfaces with infinite, discrete automorphism group. These lead to fiber bundles over the punctured disc $\mathbb{D}^\circ$ that are locally Moishezon but not globally Moishezon (using Proposition 3.22).

4.4 Singular Kähler Moishezon space need not be projective

By blowing down elliptic curves, such an easy example is not possible. Instead, consider a cubic $C \subset \mathbb{P}_2$ and let $x_1, \ldots, x_{10}$ be general points on C. Let $f : X \to \mathbb{P}_2$ be the blow-up of these points. Then the strict transform $\hat{C}$ of C in X is elliptic with $\hat{C}^2 = -1$. It can be shown that the blow-down of $\hat{C}$ is not projective.

4.5 Fiberwise projective morphism need not be a projective morphism

Let $S_0 := (g = 0) \subset \mathbb{P}^3_{\mathrm{x}}$ and $S_1 := (f = 0) \subset \mathbb{P}^3_{\mathrm{x}}$ be surfaces of the same degree. Assume that S_0 has only ordinary nodes, S_1 is smooth $\operatorname{Pic}(S_1)$ is generated by the restriction of $\mathcal{O}_{\mathbb{P}^3}(1)$ and S_1

does not contain any of the singular points of S_0. Fix $m \geq 2$ and consider

$$X_m := (g - t^m f = 0) \subset \mathbb{P}^1_{\mathbf{x}} \times \mathbb{A}^1_t.$$

The singularities are locally analytically of the form $xy+z^2-t^m=0$. Thus X_m is locally analytically factorial if m is odd. If m is even then X_m is factorial since the general fiber has Picard number 1, but it is not locally analytically factorial; blowing up $\big(x = z - t^{m/2} = 0\big)$ gives a small resolution. Thus we get that (4.1) X_m is bimeromorphic to a proper, smooth family of projective surfaces iff m is even, but (4.2) X_m is not bimeromorphic to a smooth, projective family of surfaces.

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